RetailCo: Case Analysis and Strategy Evaluation

From diagnosis to recommendation, and from recommendation to the system this programme builds

> Read after you have prepared the case and discussed it in class. Every number below is computed from the case exhibits by a model that is reproduced in the tables; where the analysis needs an assumption, it is marked as one. Industry evidence is cited to its source.

1. Executive summary

Recommendation. RetailCo should build its own AI agent (option D), launched as a staged pilot on web chat with Fiona Okafor's five conditions written in as acceptance criteria; fix the root cause of the volume in parallel by reviewing the carrier mix; bridge the coming peak with temporary staff; and commit publicly to the floor that the agent takes over repetitive work, not jobs. RetailCo should reject outsourcing (B), whose three-year contract removes the option to change course, and decline the hosted platforms (E), which at mid-range prices cost more per resolved contact than RetailCo's own people.

Finding	Number	Where
Contacts that exist only because contacts per order rose from 1.1 to 1.6	**4,660 a day, 31%**	§2
Net annual effect of the 2023 carrier change once guarantee and support costs are counted	**−£1.43M**	§3
Steady-state net annual benefit of D on chat (75% containment, £0.15 per contact, £0.20M run team)	**£0.94M**	§5.2
Three-year NPV at 10%: D versus B	**£2.13M versus £1.82M**	§5.3
Weighted score out of 5: D (next best A at 3.05)	**4.15**	§5.4
Containment at which D breaks even, steady state / in the build year	**30% / 47%**	§5.6
Expected value of D as a staged investment (maximum loss £0.35M)	**£1.42M**	§5.5

2. Diagnosis: what is the real problem?

The board sees four symptoms: cost growing 22% a year, a 4.2-hour first reply, CSAT at 61% and attrition at 35%. Treated separately, each suggests a different fix: hire for response time, outsource for cost, automate for both. Treated as a system, they share causes.

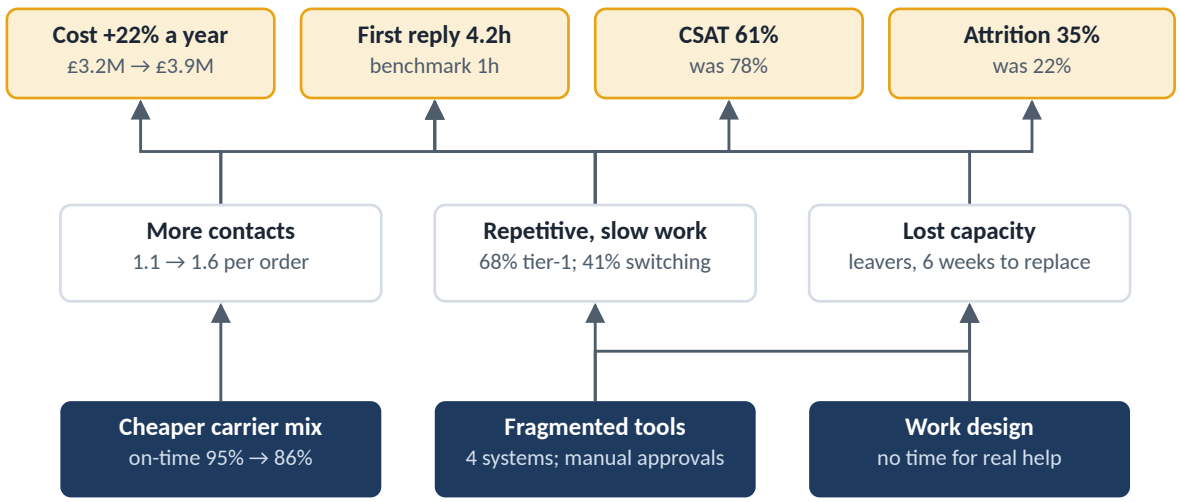


Figure A1 Problem tree. Root causes (navy) drive mechanisms (white) that produce the symptoms on the board's agenda (amber). Arrows point from cause to effect, read bottom-up.

The decisive number is in Exhibit 2. RetailCo ships about 9,315 orders a day. At 1.1 contacts per order, that would generate about 10,250 contacts; at 1.6 it generates about 14,900. Roughly 4,660 contacts a day, 31% of all volume, exist because customers are chasing parcels and refunds that the old carrier arrangement did not create.

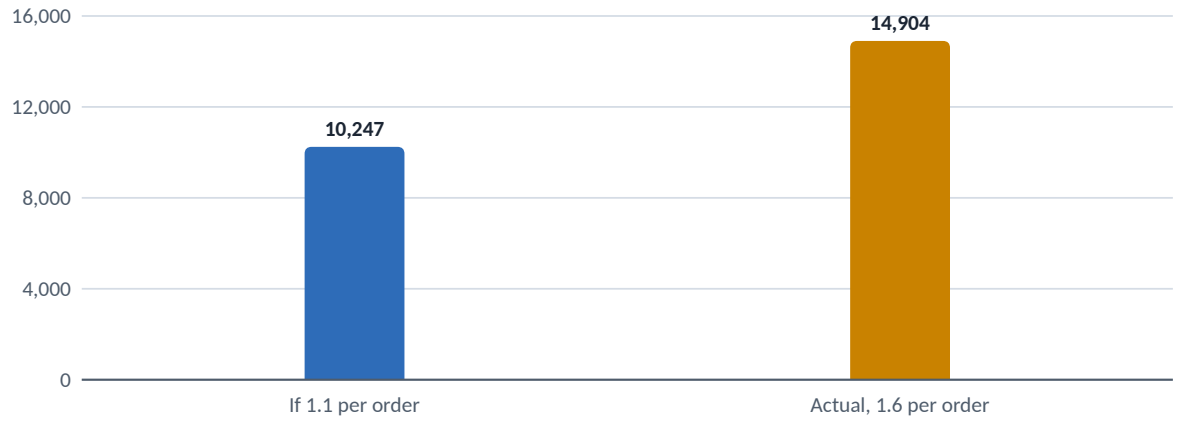


Figure A3 Contacts per day. About 4,658 a day (31%) exist only because contacts per order rose.

Table A1 · Volume decomposition	Value	Calculation
Orders per day	9,315	3.4M ÷ 365
Contacts per day at 1.1 per order	10,247	9,315 × 1.1
Contacts per day at 1.6 per order	14,904	9,315 × 1.6 (matches the case's 15,000)
Excess contacts per day	**4,658 (31%)**	difference
Annual cost of handling the excess	**£1.00M**	4,658 × 81% reaching an agent × 365 × £0.72

The symptoms also feed one another. Late parcels create contacts; contacts create a backlog; the backlog creates repeat contacts ("third time I've asked"); repeat contacts and repetitive work wear agents down; agents leave; capacity falls; the backlog grows. This reinforcing loop explains why the numbers in Exhibit 7 accelerate rather than drift.

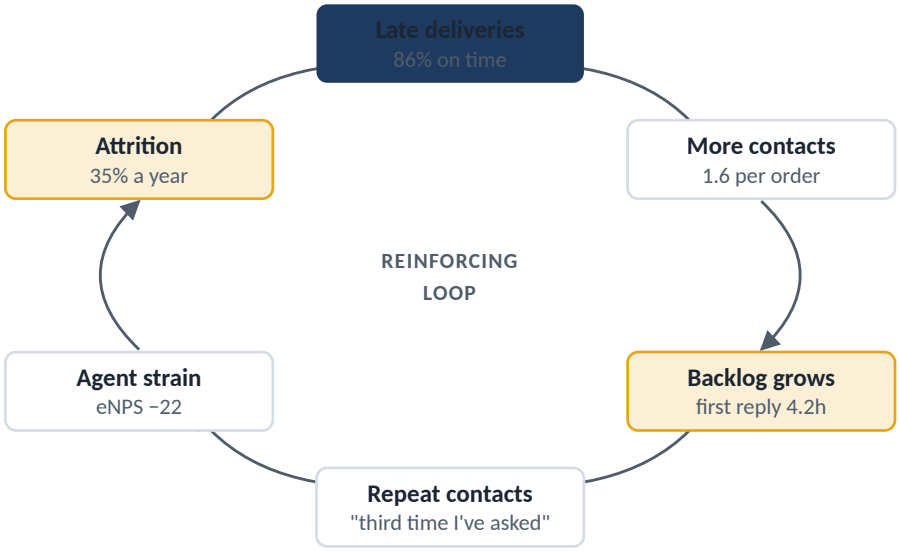


Figure A2 The reinforcing loop. Without intervention each turn makes the next worse; automation breaks the loop at 'more contacts' and 'agent strain', logistics at 'late deliveries'.

Implication. RetailCo has a volume problem caused by operations, made worse by a work-design problem inside support. Any recommendation that answers only one of these leaves the

loop running. Automation can absorb the volume and change the work; only logistics can remove the excess volume at source.

3. The carrier decision, revisited

The 2023 carrier change was evaluated as a saving of about £0.70 per parcel, £2.38M a year. Its full cost appeared elsewhere in the P&L: £2.81M a year in guarantee refunds and credits (Exhibit 10) and about £1.00M a year in extra support contacts (Table A1). Counted together, the change costs RetailCo about £1.43M a year, before any effect of lower satisfaction on repeat purchase.

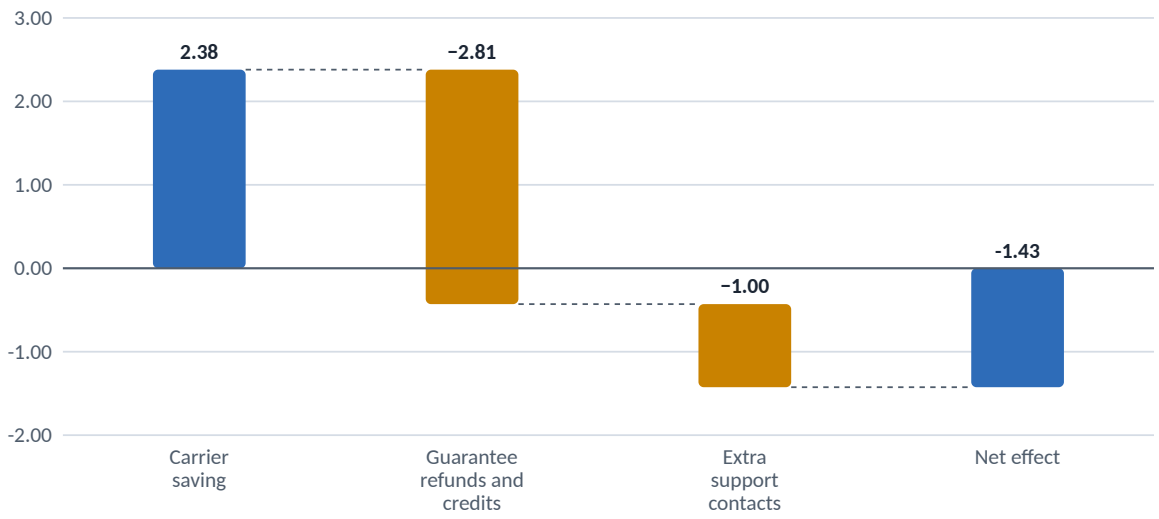


Figure A4 The 2023 carrier change, annualised, £M. A saving of £2.38M becomes a net loss of about £1.43M before any lost sales.

Table A2 · Carrier change economics, annualised	£M
Delivery cost saving (£0.70 × 3.4M parcels)	+2.38
Late Delivery Guarantee refunds and credits	−2.81
Handling the excess contacts	−1.00
Net effect	**−1.43**
Cost of each 1 percentage point of orders arriving 7+ days late	1.80 (34,000 refunds × £53)

This is a classic case of a decision optimised within one function, here delivery cost per parcel, whose costs landed in two others, customer operations and marketing. Priya's board paper

should say so plainly but without blame, and should bring Rob Ellis, the COO who owns the carrier contracts, into the recommendation. Moving volume away from Carrier C, which delivers only 79% of parcels on time, is likely the single highest-return action available. The case does not give enough data to size it precisely; Priya should ask for parcel-level lateness by carrier before the board meeting.

4. Stakeholders



Figure A5 Stakeholder map. Rob Ellis is the overlooked stakeholder: he owns the root cause. Tom Barker's low formal power hides high practical power over adoption.

Table A3 · Stakeholder	**Interest**	**Main concern**	**Needs to see**	**Likely position**
Andrew Whitfield, CEO	Recovery before next peak; no embarrassment	Public failure of an AI system	One recommendation; controls; stop conditions	Open to D if risk is visibly managed
Martin Hale, CFO	Cost growth stopped; payback	Sunk cost with nothing to show	NPV, staged spend, exit plan	Leans B; persuadable by option value
Fiona Okafor, Risk	Compliance, liability, data	Unapproved money or promises; opacity	Five conditions met in design and tested	Supports D or B; opposes E as described
Daniel Reyes, CTO	Building capability	Defining "good enough"	Agreed targets and evaluation before build	Champions D
Claire Dunmore, CCO	Brand and customer trust	Customers' dislike of AI (Gartner: 64%)	Easy route to a person; disclosure; quality	Cautious; supports a pilot
Rob Ellis, COO	Delivery cost	Reversing a saving he delivered	Full economics of the carrier mix	Not yet engaged: the overlooked stakeholder
Tom Barker and the floor	Jobs, meaningful work	Automation as a euphemism for cuts	A written commitment; a role in the design	Sceptical after 2021; can make or break adoption
Customers	Fast, correct answers; a person when needed	Being trapped with a bot	Answers in seconds; a clear way out	Indifferent to the method if service is good

5. Strategy evaluation

5.1 Criteria

The criteria come from the case, not from a textbook: Whitfield's note (cost, recovery before peak, no embarrassment), Hale's demand for payback and an exit, Okafor's conditions, Barker's question about jobs, and Reyes's uncertainty about delivery. Weights reflect the CEO's priorities and are tested for robustness in §5.4.

Table A4 · Criterion	Weight	Source in the case
Economics (net annual effect, NPV)	25%	Whitfield, Hale
Service recovery before next peak	20%	Whitfield; Exhibit 7
Control and compliance	20%	Okafor memo (Exhibit 17); *Moffatt v. Air Canada*
People and attrition	15%	Barker; staff survey (Exhibit 11)
Reversibility and optionality	10%	Hale's "what happens if it doesn't work"
Delivery risk	10%	Reyes; no prior AI delivery

5.2 Economics by option

All options are compared with the status quo, in steady state, using the case's own figures. The key assumptions are listed so they can be challenged.

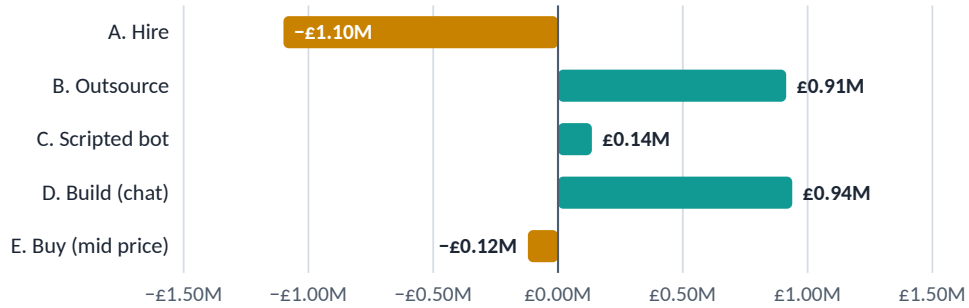


Figure A6 Steady-state net annual effect against the status quo, £M (positive is better). D includes £0.20M a year for a two-engineer run team.

Table A5 · Option	Steady-state net effect (£M/yr)	How it is calculated	Key assumption
A. Hire	−1.10	Extra cost from Exhibit 15	Service recovers in ~5 months; work unchanged
B. Outsource	+0.91	3.03M tier-1 agent contacts × (£0.72 − £0.38) − 10% retained oversight	Quality holds; savings realised as staff leave (no redundancies)
C. Scripted bot	+0.14	15% of chat resolved × £0.72 − £0.18M licence	15% resolution, consistent with the 2021 experience
D. Build (chat)	+0.94	2.92M chats: 75% × £0.72 avoided (£1.58M) − £0.15 × all chats (£0.44M) − £0.20M run team	75% containment; £0.15 per contact
D, adding email in year 2	+0.35 more	1.10M emails: 65% × £0.72 − £0.15 × all	65% containment on email
E. Buy platform	−0.12 (range −0.50 to +0.26)	2.19M resolutions × (£0.72 − price £0.60–£0.95)	Same 75% containment as D

The platform paradox. Option E is priced per resolution at £0.60 to £0.95. RetailCo′s own cost per agent-handled contact is £0.72. At the midpoint of £0.775, every contact the platform resolves costs more than a person would. E saves money only at the bottom of its price range, and even then less than D. This is the single most frequently missed number in class discussion.

B and D are close on steady-state economics. At £0.91M and £0.94M a year, outsourcing and building are within a few percent of each other. The case for D therefore rests less on cost than on the other criteria: reversibility, control, the effect on staff, and the option to extend to email and to other uses.

5.3 Three-year net present value

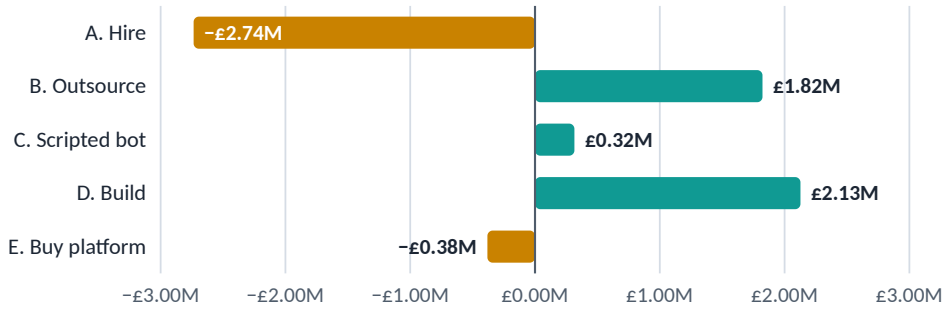


Figure A7 Three-year net present value at a 10% discount rate, £M, including build, transition and ramp-up.

Table A6 · Cash flows (£M) vs status quo	Year 1	Year 2	Year 3	NPV at 10%
A. Hire	−1.10	−1.10	−1.10	**−2.74**
B. Outsource (4-month transition, half double-running)	+0.42	+0.91	+0.91	**+1.82**
C. Scripted bot (2 months to launch)	+0.11	+0.14	+0.14	**+0.32**
D. Build (£0.35M build; benefits from month 5; email from year 2)	+0.28	+1.10	+1.29	**+2.13**
E. Buy (assumed £0.15M integration; 6 months to launch)	−0.21	−0.12	−0.12	**−0.38**

5.4 Weighted scoring and robustness

Table A7 · Score (1–5)	Econ. 25%	Service 20%	Control 20%	People 15%	Reversible 10%	Delivery 10%	Weighted
A. Hire	1	4	5	2	3	4	3.05
B. Outsource	4	3	3	1	1	3	2.75
C. Scripted bot	2	2	4	2	5	5	3.00
D. Build	5	4	4	4	5	2	**4.15**
E. Buy platform	2	4	2	4	3	3	2.90

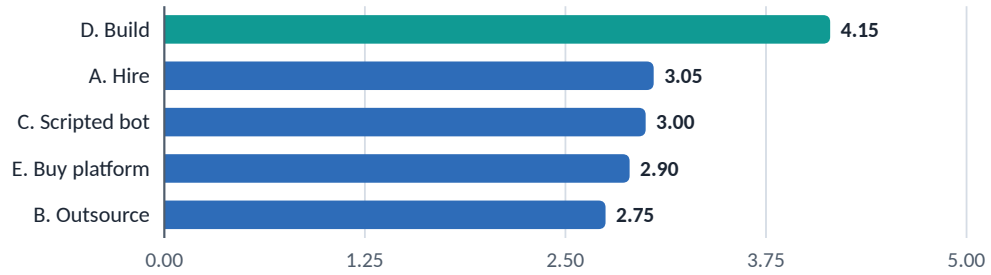


Figure A8 Weighted score out of 5. Under 729 combinations of each weight at 50%, 100% and 150% of base, D ranks first in 100% of cases.

Scoring matrices are only as good as their weights, so the weights were stress-tested: each was set independently to 50%, 100% and 150% of its base value, giving 729 combinations. D ranked first in all of them. D's weakness, delivery risk, is real but is the one criterion a staged pilot directly reduces.

5.5 D as a real option

Martin Hale's fear is a sunk £0.35M with nothing to show. Framed correctly, D is not a £0.35M bet on an outcome but the purchase of an option: a twelve-week build and a pilot that reveals the containment rate, after which RetailCo chooses to scale, narrow or stop. B offers no such option; its three-year contract commits the company before it learns anything.

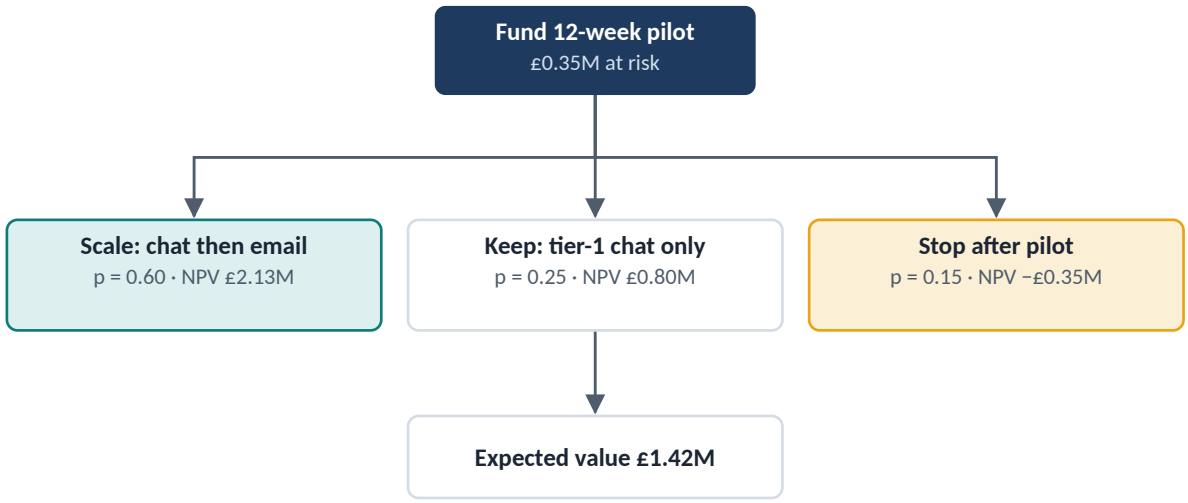


Figure A11 Option D as a staged investment. Probabilities are the analyst's assumptions, to be replaced with pilot evidence; the maximum loss is the pilot cost.

With assumed probabilities of 60% (scale), 25% (narrow to tier-1 chat) and 15% (stop), the expected value is about £1.42M, against a maximum loss of £0.35M. The probabilities are assumptions; the structure is the point. The pilot replaces them with evidence within sixteen weeks.

5.6 Break-even and sensitivity

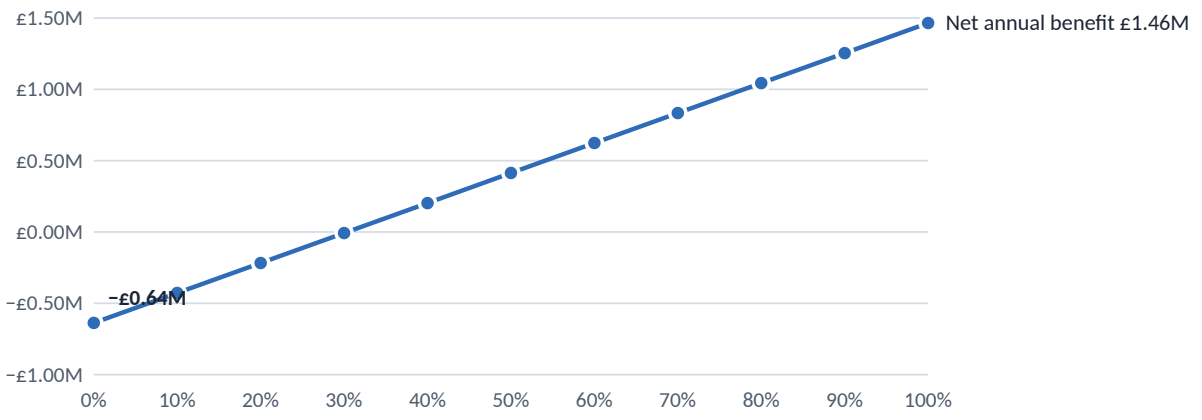


Figure A9 Steady-state net annual benefit of D (chat) against containment, £M. Break-even is about 30% in steady state and 47% in a year that carries the build cost.

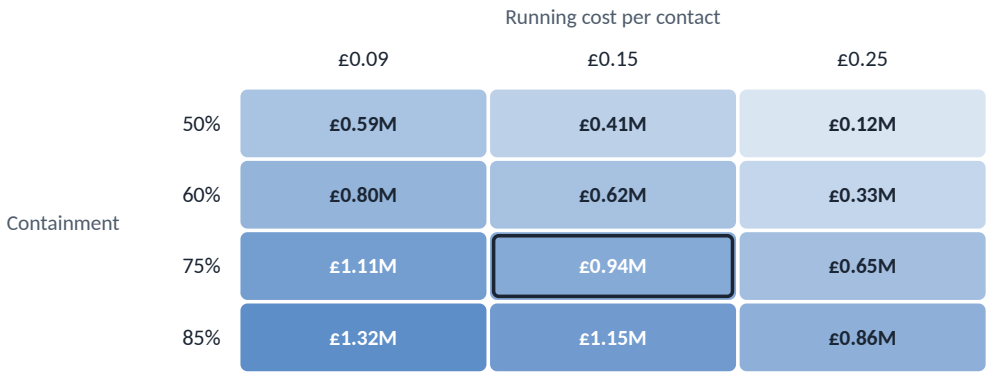


Figure A10 Sensitivity of D's steady-state net annual benefit, £M. Outlined: the base case. Every cell is positive; containment matters more than running cost.

D breaks even at about 30% containment in steady state, and about 47% in a year that carries the full build cost. Published deployments make 75% a reasonable planning case: Klarna reported its assistant handling about two-thirds of service chats in its first month (Exhibit 14), and Gartner has since predicted that agentic AI will resolve 80% of common customer service issues without human intervention by 2029. RetailCo′s mix is favourable: 68% of contacts are rule-based tier-1 work. Containment, not running cost, is the variable that matters, so the pilot′s first job is to measure it honestly.

6. Risks

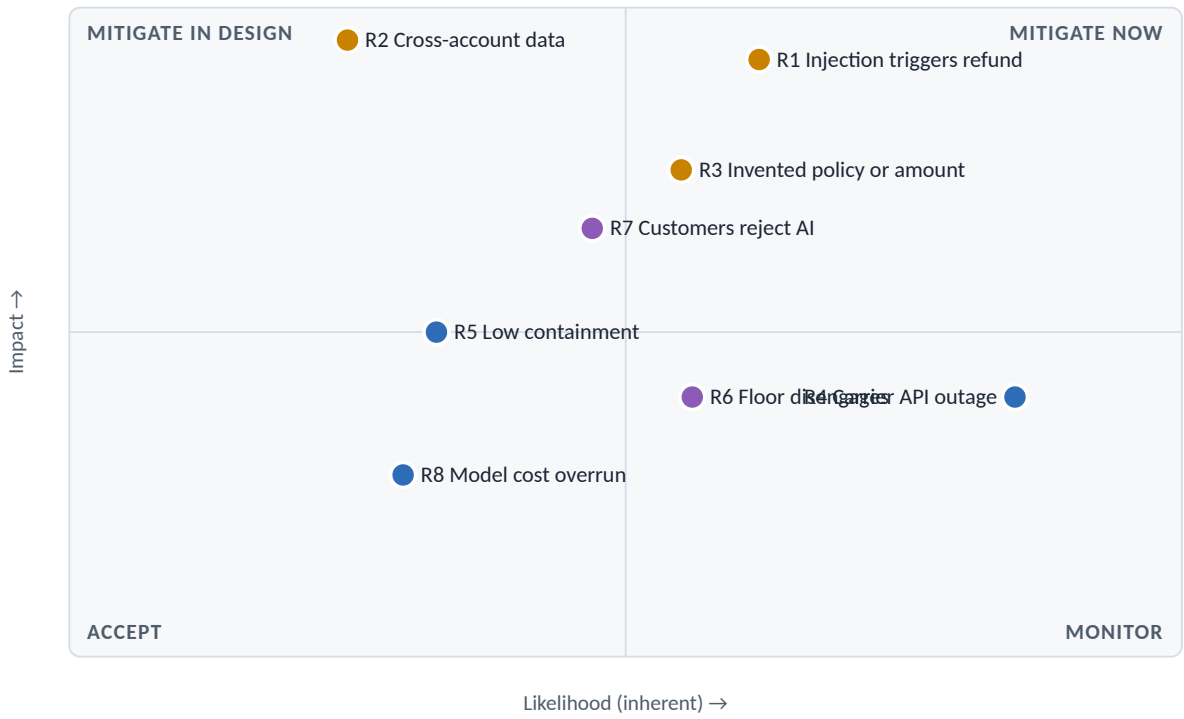


Figure A12 Inherent risk before controls. Amber: safety and legal; blue: operational and economic; purple: people and customers.

Table A8 · Risk	Inherent	Control (and where it is built)	Residual
R1 Prompt injection triggers a refund	High	Refund limit enforced in code twice: inside the refund tool and in a pre-action guard; approval queue above £200 (Modules 5, 9, 10)	Low
R2 Customer sees another's data	High impact	Identity from the verified session; ownership checked inside every tool; identical "not found" responses (Modules 4, 5, 10)	Very low
R3 Agent invents a policy or amount	High	Policy only from retrieved articles; mechanical grounding check on every amount and reference (Modules 7, 11)	Low
R4 Carrier API outage	Likely	Retries with back-off, cache with "last known status", circuit breaker (Modules 9, 10)	Low
R5 Containment below plan	Medium	Golden set before build; shadow mode; staged ramp; stop condition (Modules 1, 13, 14)	Medium, bounded at £0.35M
R6 Floor disengages	Medium	No-redundancy commitment; agents write the golden set and staff approvals (Module 14; people plan)	Low–medium
R7 Customers reject AI	Medium	Disclosure; one-step route to a person; quality targets (Modules 11, 12)	Medium
R8 Model cost overrun	Low	Model tiering; cost per contact tracked from traces (Modules 3, 12, 14)	Low

7. Recommendation

1. **Build the agent for web chat (option D),** with Fiona Okafor's five conditions as written acceptance criteria, a golden set agreed before code, and two go/no-go points: after shadow mode (week 12) and after the 5% canary (week 14).
2. **Fix the root cause.** Commission the COO to review the carrier mix within eight weeks, starting with Carrier C. Report lateness and guarantee cost by carrier monthly.

3. **Bridge the peak** with temporary agents from November to January, sized to keep first response under two hours. This is a small, reversible use of option A.
4. **Commit to the floor in writing:** no compulsory redundancies for twelve months; senior agents join the build to write golden cases, staff the approvals queue and handle escalations; new roles for quality analysts who read traces.
5. **Do not sign a multi-year outsourcing contract now.** Revisit B for email or phone overflow only if D′s pilot fails.
6. **Use E as a price benchmark,** not a purchase, unless a vendor offers per-resolution pricing well below £0.72 with full traceability and data processed in RetailCo′s environment.

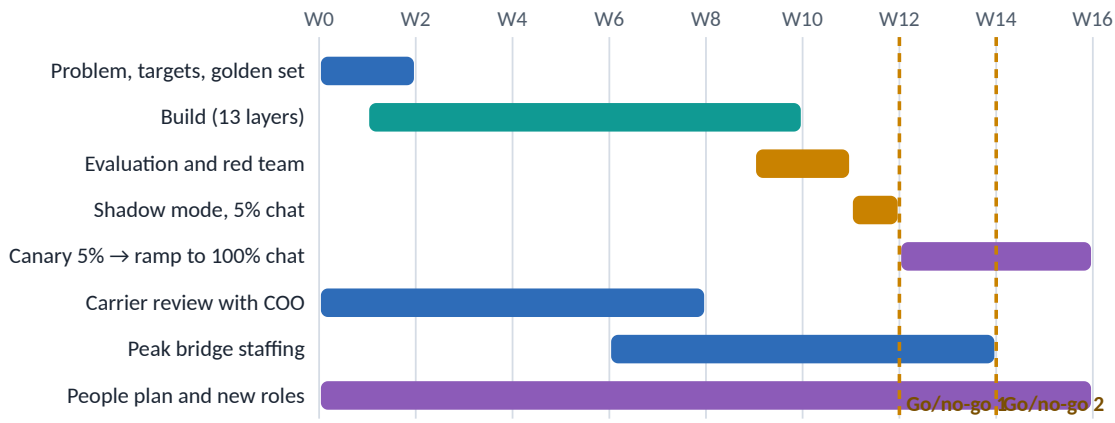


Figure A13 Integrated 16-week plan: the agent, the root-cause fix and the people plan run together.

8. Success measures and stop conditions

Table A9 · Measure	Target	Baseline	Source	Owner
Containment (resolved, no repeat within 48h)	> 75%	0% (all chats reach a person)	Escalation table; repeat-contact join	Priya Raman
CSAT, chat	> 80%	61%	Post-contact survey	Claire Dunmore
First response, tier-1 chat	< 30 s	4.2 h	Trace durations	Daniel Reyes
Cost per contact	< £0.15	£0.72	Token usage per trace × price list	Martin Hale
Critical safety incidents	0	—	Incident log; guard denials	Fiona Okafor
Guard rail: repeat contacts within 48h	Not above baseline	Measure in shadow mode	Contact history	Priya Raman
Guard rail: refunds reversed or disputed	Not above baseline	Current rate	Payments	Martin Hale
Stop at once: any refund outside policy, or any cross-account exposure	Kill switch; route chat to people; review before restart			
Pause the ramp: CSAT below 61% for 7 days, or containment below 60% after 4 weeks	Error analysis on traces; fix; repeat the stage			

9. From analysis to architecture

The analysis produces requirements. Each requirement maps to one or more of the thirteen layers, to the module in this programme where it is built, and to the test or golden case that proves it. This table is the spine of the programme: every module answers a question first asked in the case.

Table A10 · Requirement (source)	**Layers**	**Built in**	**Evidence in the repository**
R1 Handle tier-1 chat end to end (Exhibit 5)	L1, L3, L6, L7	Modules 4, 6, 8, 9	Golden cases g01–g05, g11, g12
R2 Apply the Late Delivery Guarantee exactly (Exhibit 10)	L9	Module 10	`test_eligibility_rules` ; g03
R3 Refunds above £200 need a person (Okafor 2)	L2, L7, L9	Modules 5, 9, 10	`test_refund_is_idempotent_and_capped` , `test_refund_cap_enforced…` , `test_high_value_refund_goes_to_approval` ; g04
R4 No cross-account data (Okafor 1)	L1, L2, L9	Modules 4, 5, 10	`test_cross_account_session_is_blocked` , `test_ownership_enforced_in_code` ; g09
R5 Every decision traceable (Okafor 3)	L6, L13	Modules 8, 12	`test_trace_has_a_span_per_layer`
R6 Tell customers it is an AI (Okafor 4)	L12	Module 11	`test_ai_disclosure_on_first_reply`
R7 Stop all actions at once (Okafor 5)	L2, L9, L13	Modules 5, 12	`test_kill_switch_blocks_writes` , `test_kill_switch`
R8 Never refund twice (Hale)	L8, L9	Module 9	`test_late_refund_full_flow`
R9 Survive carrier outages (Exhibit 9: 99.1%)	L8, L9	Modules 9, 10	`test_carrier_falls_back_to_cache` , `test_queue_retries…`
R10 Partial returns go to people (Exhibit 10)	L3	Module 6	g07
R11 Policy answers only from approved articles (Air Canada)	L4, L11	Modules 7, 11	`test_policy_question_answered_from_kb` ; g08

Table A10 · Requirement (source)	Layers	Built in	Evidence in the repository
R12 Resist manipulation (Dec 2023 chatbot incident)	L2, L6	Module 5	`test_injection_escalates`; g06; red team
R13 Response < 30 s at < £0.15 (targets)	L5, L13	Modules 3, 8, 12, 14	Trace durations; cost model
R14 Know when it is good enough (Reyes)	L13	Modules 1, 13	Golden set; CI evals; LLM judge
R15 People keep the hard cases (Barker)	L6, L9, L13	Modules 8, 12, 14	Escalation and approval queues

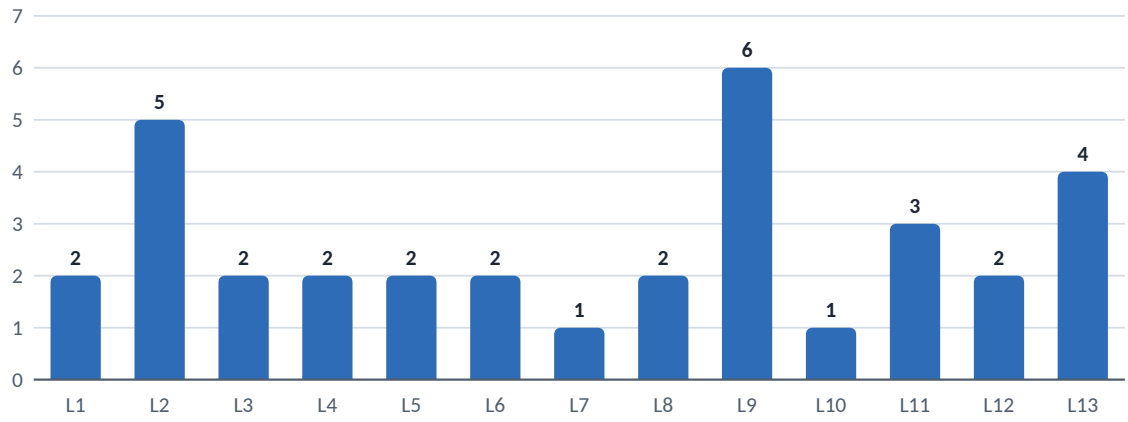


Figure A14 Number of case requirements (Table A9) that each layer helps meet. Tools, safety and monitoring carry the most; the model layers carry the fewest.

10. What has happened since

The case is set in September 2024. Since then the evidence has moved further towards automation with controls. In March 2025 Gartner predicted that by 2029 agentic AI would autonomously resolve 80% of common customer service issues without human intervention, leading to a 30% reduction in operational costs. Online retail's share has continued to edge up: the ONS reported internet sales at 27.1% of GB retail sales for 2024, and 28.3% in September 2026. The research base has also grown: Dell'Acqua and colleagues' field experiment with consultants (Harvard Business School working paper 24-013, 2023) found large gains on tasks "inside the frontier" of AI capability (12.2% more tasks, 25.1% faster, markedly higher quality)

and lower accuracy on tasks outside it, which is the case for routing by task type that RetailCo's decision gate implements. Noy and Zhang (*Science*, 2023) found writing tasks took 40% less time with 18% higher quality.

Table A11 · Development	Date	Relevance to RetailCo
Gartner: agentic AI to resolve 80% of common issues by 2029; 30% lower operating cost	Mar 2025	Supports a 75% planning case for tier-1
ONS: internet share of GB retail 27.1% (2024); 28.3% (Sep 2026)	2025–26	Online service volume keeps growing
Dell'Acqua et al., "jagged frontier" field experiment	2023	Route by task: automate inside the frontier, escalate outside it
Noy and Zhang, *Science*	2023	Large gains on writing tasks such as replies

11. Limits of this analysis

- **Containment is unknown until measured.** Every economic conclusion rests on it; the staged pilot exists to measure it.
- **Savings arrive through attrition.** With no redundancies, cost falls as about 36 people a year leave and are not replaced. The cash benefit lags the operational one.
- **The hardest contacts remain.** The 25% that still reach people are longer and harder, so cost per human-handled contact will rise even as total cost falls.
- **The carrier review may change everything.** If contacts per order return to 1.1, volume falls by about 31% and the scale of any automation should be revisited.
- **Customer attitudes vary.** Gartner's 64% figure describes stated preferences; observed behaviour when service is fast and a person is one step away may differ. The pilot should measure it.

References

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